

Support Ticket Classification and Prioritization

Using NLP and Machine Learning

Objective

- - Classify tickets into categories (e.g., Fileservice, Software, etc.)
- - Assign priority levels (High, Medium, Low)
- - Improve support operations by automating ticket sorting

Import Libraries and Setup

- - Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn, nltk, joblib
- - NLTK data downloaded: stopwords, wordnet
- - Warnings suppressed for cleaner output

Load Support Ticket Dataset

- - Loaded X_train.csv, y_train.csv, X_test.csv, y_test.csv
- - Merged X and y for train and test sets
- - Combined into full dataset
- - Dataset shape and columns displayed

Data Preprocessing

- - Text cleaning: lowercase, remove punctuation and numbers, extra whitespace
- - Tokenization, stopwords removal, lemmatization
- - Applied to 'text' column to create 'cleaned_text'

Exploratory Data Analysis (EDA)

- - Category distribution: bar plot of category_truth
- - Text length analysis: statistics and histogram
- - Priority mapping: categories to High/Medium/Low

Feature Engineering

- - TF-IDF Vectorization: max_features=5000, ngram_range=(1,2)
- - Transformed cleaned_text into feature vectors
- - Prepared X_train_vec, X_test_vec for training

Model Training

- - Category Classification: Logistic Regression
- - Priority Prediction: Multinomial Naive Bayes
- - Trained on vectorized text data

Model Evaluation

- - Category: Classification report and confusion matrix
- - Priority: Classification report and confusion matrix
- - Visualized using seaborn heatmaps

Model Deployment Preparation

- - Saved models: `category_model.pkl`, `priority_model.pkl`
- - Saved vectorizer: `tfidf_vectorizer.pkl`
- - Inference function: `classify_ticket()` for predictions

Conclusion

- - Automates support ticket classification and prioritization
- - High accuracy for category classification
- - Priority based on business logic
- - Benefits: faster response, better customer satisfaction
- - Ready for integration into ticketing systems